

Construction of Galactic Stellar Orbits Similar to Harmonic Oscillators. I

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Starting from a two-dimensional harmonic oscillator with its own configuration space, it is possible to construct a stellar orbit which traces a curve in the moving galactic meridian plane topologically equivalent to a Lissajous curve. This is done by introducing a non-Cartesian metric in the configuration space of the harmonic oscillator and then applying to the coordinates and the time a topology-preserving canonical transformation (or similarity transformation). The transformation functions define the resulting galactic potential function along the stellar orbit. The problem of matching this potential function exactly with a given galactic potential function remains unsolved. Examples of orbits constructed in this way are given and compared with previously obtained numerical stellar orbits in a model of the Galaxy.

OBJECTIVES

DURING the past few years numerous numerical integrations of the equations of motion for stars in model galaxies have been carried out. They have indicated the existence of a class of orbits which trace curves in the moving galactic meridian plane topologically equivalent to Lissajous figures. [See Ollongren (1962, 1965), Contopoulos (1957, 1960), and Contopoulos and Woltjer (1964) for examples.] The above-mentioned classification of orbits is *geometrical* in nature. Attempts at establishing a *dynamical connection* between harmonic oscillators and galactic stellar orbits have been undertaken by Ollongren (1962) and by van de Hulst (1962). As a starting point a harmonic oscillator was considered; it was attempted to distort the corresponding Lissajous curve continuously and to apply a time transformation in such a manner that a galactic stellar orbit was obtained. The results were perhaps not so useful mainly because the resulting galactic potential function was expressed in terms of the configuration coordinates of the harmonic oscillator, rather than in the galactic ones.

Recently the problem was reformulated and embedded in the more general theory of topology-preserving, completely canonical mappings in configuration space (Ollongren 1966). As a result a more useful theory is obtained, and the necessary and sufficient conditions for a galactic potential admitting of orbits similar to harmonic oscillators is found in terms of the galactic configuration coordinates. In the next section a resume of the theory is given.

The theory can be used for *constructing* orbits similar to harmonic oscillators and having certain prescribed properties of galactic stellar orbits. It is the purpose of this paper to discuss some aspects of such constructions; a particular form is chosen and it is shown how well previously numerically computed galactic orbits match with constructed orbits.

THEORY

Under the usual assumptions of stellar dynamics the orbit of an individual star is completely determined by the time-independent gravitational attraction of the stellar system as a whole; the (conservative) field of force has an axis and a plane of symmetry. At each instant a plane (*the meridian plane*) is defined by the position of the star in 3 space, and the axis of symmetry. It is well known that the motion of the star in the (moving) meridian plane can be described by the Hamiltonian

$$\mathcal{H}_s = \frac{1}{2}p_\varpi^2 + \frac{1}{2}p_z^2 + U(\varpi, z), \quad (1)$$

in which ϖ and z are rectangular Cartesian coordinates taken in the usual way: z along the axis of symmetry, and ϖ in the galactic plane. The function U , called the *reduced potential function*, depends on the distribution of mass in the stellar system and the angular momentum integral of the star considered. It has a minimum at $\varpi = \varpi_e$, $z = 0$. This point corresponds to an equilibrium solution of above Hamiltonian system (circular motion in 3 space in the plane of symmetry) with energy $E_e = U_e(\varpi_e, 0)$. Also, $U(\varpi, z) = U(\varpi, -z)$. We are concerned with motion in a neighborhood of the equilibrium solution; the size is of the order of several kpc² on the galactic scale. In this neighborhood we assume that U is a smooth function of both arguments (i.e., twice continuously differentiable) and that it increases monotonically along any half straight line originating at the point of equilibrium; also it increases monotonically along any half straight line perpendicular to and with its end point on the ϖ axis. So we have here a locally smooth potential well. In order to formulate the conditions which U must satisfy in order that there exist orbits similar to harmonic oscillators we proceed as follows:

Consider a harmonic oscillator in a configuration space with nondimensional rectangular Cartesian coordinates x_1 and x_2 ; let it be centered on the point $(c_2, 0)$ on the x_1 axis, and let it be symmetric with respect to that axis. In terms of x_1 and x_2 the Lagrangian function

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is then

$$\mathcal{L} = \frac{1}{2}\dot{x}_1^2 + \frac{1}{2}\dot{x}_2^2 - V(x_1, x_2), \quad (2)$$

with

$$V = \frac{1}{2}a(x_1 - c_2)^2 + \frac{1}{2}bx_2^2 \quad (3)$$

as the potential function. The dot denotes differentiation with respect to the nondimensional time t . The ratio of the oscillation frequencies in the x_1 -resp. x_2 coordinates is denoted by γ and

$$\gamma = (a/b)^{1/2}.$$

General curvilinear coordinates (q_1, q_2) are introduced in the (x_1, x_2) plane by the transformation

$$x_1 = x_1(q_1, q_2) \quad x_2 = x_2(q_1, q_2),$$

assumed to be continuously differentiable with a nonvanishing Jacobian. The line element in the configuration space is in terms of the new coordinates

$$ds^2 = \sum_{i,j} a_{ij} dq_i dq_j$$

with

$$a_{ij} = \frac{\partial x_1}{\partial q_i} \frac{\partial x_1}{\partial q_j} + \frac{\partial x_2}{\partial q_i} \frac{\partial x_2}{\partial q_j}. \quad (4)$$

(The indices i and j both take the values 1 and 2 throughout this section.)

The Lagrangian function becomes

$$\mathcal{L} = \frac{1}{2} \sum_{i,j} a_{ij} \dot{q}_i \dot{q}_j - V(q_1, q_2).$$

It can be shown that the Hamiltonian function becomes

$$\mathcal{H}_h = \frac{1}{2} \sum_{i,j} b_{ij} p_i p_j + V(q_1, q_2), \quad (5)$$

in which the matrix (b_{ij}) is the inverse of the matrix (a_{ij}) and

$$p_i = \partial \mathcal{L} / \partial \dot{q}_i.$$

Next the Hamiltonian (5) is subjected to a coordinate transformation and a time transformation:

$$q_i = f_i(Q_1, Q_2), \quad (6.1)$$

$$P_i = \alpha_{1i} p_1 + \alpha_{2i} p_2, \quad (6.2)$$

and

$$dt = g(Q_1, Q_2) dT. \quad (6.3)$$

Here the functions f_1 and f_2 are assumed to be continuously differentiable, g is a continuous function, T is the transformed time, and

$$\alpha_{ij} = \partial f_j / \partial Q_i.$$

The transformation (6.1), (6.2) is completely canonical if and only if the functional determinant $|\alpha_{ij}|$ is equal to unity. Such a transformation can be called a *similarity transformation* since the solution curves in the (q_1, q_2)

space are topologically equivalent to those in the (Q_1, Q_2) space, or *similar* in this sense.

The new Hamiltonian can be found in terms of the new coordinates, momenta and time, by considering the *extended phase space*; see Szebehely (1967) and Ollongren (1966). In the latter work the conditions are formulated under which the new Hamiltonian takes the form

$$\mathcal{H}^* = \frac{1}{2}P_1^2 + \frac{1}{2}P_2^2 + V^*(Q_1, Q_2). \quad (7)$$

Without derivation we state the result:

Given a metric tensor a_{ij} , the functions f_1 and f_2 must satisfy the following system of partial differential equations

$$\begin{aligned} g(\alpha_{11}^2 + \alpha_{21}^2) &= a_{22}, \\ g(\alpha_{11}\alpha_{12} + \alpha_{22}\alpha_{21}) &= -a_{12}, \\ g(\alpha_{12}^2 + \alpha_{22}^2) &= a_{11}, \end{aligned} \quad (8.1)$$

where

$$g = (a_{11}a_{22} - a_{12}^2)^{1/2}. \quad (8.2)$$

The function V^* takes the form

$$V^* = g(Q_1, Q_2) \{ V(Q_1, Q_2) - E \} + E,$$

in which E is the numerical value of the energy of the harmonic oscillator and

$$V = \frac{1}{2}a[x_1\{f_1(Q_1, Q_2), f_2(Q_1, Q_2)\} - c_2]^2 + \frac{1}{2}bx_2\{f_1(Q_1, Q_2), f_2(Q_1, Q_2)\}^2. \quad (9)$$

Finally, if we identify Q_1 and Q_2 with dimensionless galactic coordinates ϖs^{-1} and zs^{-1} (s is a scaling constant), forms (7) and (1) show that we can identify P_1 and P_2 with dimensionless momenta $p_\varpi u^{-1}$ and $p_z u^{-1}$ (u is a scaling constant). Then $\mathcal{H}^*$ and $\mathcal{H}_h u^{-2}$ are identical provided that

$$U(\varpi, z) = u^2 g(\varpi/s, z/s) \{ V(\varpi/s, z/s) - E \} + u^2 E, \quad (10)$$

in which $V(\varpi s^{-1}, z s^{-1})$ is given by (9) with ϖs^{-1} and $z s^{-1}$ instead of Q_1 and Q_2 . Formula (10) is therefore a sufficient condition to be satisfied by a galactic potential function in order that it possesses orbits similar to those of a harmonic oscillator. This formula is used in the next sections to construct orbits of this kind.

A PARTICULAR SOLUTION

The general solution of the system of equations (8.1), (8.2) is not known. A useful particular solution is the following (see also Fig. 1):

Introduce polar coordinates (q_1, q_2) in the (x_1, x_2) plane by

$$x_1 = e^{c_1 - q_1} \cos q_2, \quad -\infty < q_1 < +\infty, \quad (11.1)$$

$$x_2 = e^{c_1 - q_1} \sin q_2 \quad -\frac{1}{2}\pi \leq q_2 \leq +\frac{1}{2}\pi.$$

The components of the metric tensor (4) are

$$a_{11} = a_{22} = e^{2(c_1 - q_1)} \quad a_{12} = 0, \quad (11.2)$$

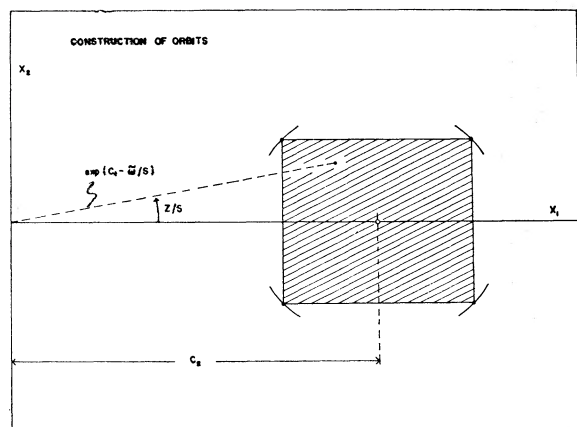


FIG. 1. Nonperiodic harmonic oscillator in its configuration space. The open circle corresponds to lowest potential energy. Polar coordinates are introduced as indicated.

and (8.2) gives

$$g = e^{2(c_1 - q_1)}. \quad (11.3)$$

The partial differential equations (8.1) are satisfied for

$$q_1 = f_1(Q_1, Q_2) = Q_1, \quad (11.4)$$

$$q_2 = f_2(Q_1, Q_2) = Q_2.$$

Putting $Q_1 = \varpi s^{-1}$ and $Q_2 = z s^{-1}$ as mentioned before (s is measured in kpc), we obtain from (10) using (9) and setting $u = 1$ temporarily

$$U(\varpi, z) = \frac{1}{2} R^2 \{ a(R \cos(z/s) - c_2)^2 + b R^2 \sin^2(z/s) - 2E \} + E \quad (12.1)$$

with

$$R = \exp(c_1 - \varpi/s). \quad (12.2)$$

CONSTRUCTIONS: PRELIMINARIES

We discuss first some preliminary restrictions for the parameters occurring in (12.1), (12.2), and give formulas for some relevant quantities.

(i) Obviously we have the conditions $c_1 > 0$, $c_2 > 0$. The function (12.1) along the ϖ axis is

$$U(\varpi, 0) = \frac{1}{2} R^2 \{ a(R - c_2)^2 - 2E \} + E$$

and

$$(\partial U / \partial \varpi)_{z=0} = 0$$

for

$$R_c = \frac{1}{4} c_2 [3 + (1 + 8\alpha^2)^{\frac{1}{2}}],$$

with the auxiliary variable α defined by

$$\alpha = + (1/c_2) (2E/a)^{\frac{1}{2}} \quad (E > 0). \quad (13)$$

Thus for the *equilibrium point* in the meridian plane

$$\varpi_c = s \{ c_1 - \log c_2 - \log [\frac{3}{4} + \frac{1}{4} (1 + 8\alpha^2)^{\frac{1}{2}}] \}. \quad (14)$$

(ii) In order that the potential function (12.1) may be an approximation to a galactic potential function,

the function must increase monotonically along any half straight line originating on the ϖ axis and perpendicular to it, in a neighborhood of the equilibrium point. Consider $z > 0$. The condition is satisfied if

$$(\partial U / \partial z)_{z=0} > 0,$$

which implies [using (12.1)]

$$a < b + a c_2 / R \cos(zs^{-1}).$$

Since this must be valid for any ϖ , we replace this condition by the stricter condition

$$a < b, \text{ or } \gamma < 1, \quad (15)$$

a condition which is satisfied for all numerical orbits discussed in Ollongren (1962).

(iii) The *geometrical center* of the Lissajous figure in the (x_1, x_2) -configuration space, is transformed into the point

$$\varpi_0 = s(c_1 - \log c_2), \quad z = 0. \quad (16)$$

Whereas it coincides with the equilibrium point for the harmonic oscillator, this is not the case for the constructed orbit [unless $\alpha = 0$, corresponding to the case $E = 0$, as seen from formula (14)].

(iv) Points of *maximum* and *minimum* ϖ for orbits with $z = 0$ and a given value of the energy are obtained from the condition

$$U(\varpi, 0) = E.$$

This gives

$$\varpi = s[c_1 - \log c_2 - \log(1 \pm \alpha)]. \quad (17)$$

Comparing with (14) and (16) it is seen that the maximum and minimum values of ϖ are situated asymmetrically with respect to both ϖ_c and ϖ_0 .

CONSTRUCTIONS: CHOICE OF PARAMETERS FOR GEOMETRY

The following parameters appear in the form of U : a and b (determining together the ratio of frequencies γ); dimensionless constants c_1 and c_2 ; scaling factors s (kpc) and u (km/sec) subject to the constraints discussed in the previous section.

We now investigate what values should be given to these constants in order that orbits with the potential function (12.1), (12.2) are approximations of galactic stellar orbits in some way. Such orbits have not been observed. So instead we try to approximate some numerically computed orbits (Ollongren 1962). We use the same galactic force function as used in that paper, and select three nonperiodic orbits close to orbits discussed there. We add the isoenergetic periodic orbit with $\gamma = 3/5$. Table I gives some data.

At the outset we require that the galactic orbit to be approximated and the constructed orbit have the same value of γ . This quantity can be found in principle for numerically computed galactic orbits with any given degree of accuracy by a method described in Ollongren

(1964). If we require in addition that $b^{\frac{1}{2}}$ is the least positive integer compatible with $\gamma = (a/b)^{\frac{1}{2}}$, being expressed in its lowest terms, then both a and b are fixed uniquely.

An orbit, similar to a harmonic oscillator, constructed by choosing (1) values of the above-mentioned parameters and (2) initial conditions, fills a region densely in the (ϖ, z) plane. The region has the shape of a box with curvilinear sides. (Such orbits are called *box orbits*.) The box has eight *fundamental points* (see Fig. 2): the two points of intersection of the curvilinear sides with the ϖ axis (points A and B), the four points on the curve $U=E$ (points C, D, C' and D'); and the points $(\varpi_c, 0)$ corresponding to equilibrium, and $(\varpi_0, 0)$ corresponding to the geometrical center of the Lissajous curve in the (x_1, x_2) plane. In addition a box orbit has two fundamental dynamical parameters: the energy and (for a periodic orbit) the period. We discuss these in the next section.

For numerically computed galactic orbits which appear to be similar to harmonic oscillators, it is possible to determine by numerical methods most of the eight fundamental points. It is our task to choose above-mentioned parameters and the initial conditions of constructed orbits in such a way that as many fundamental points as possible coincide exactly with those of actually computed galactic orbits. It is clear that we cannot make all of them coincide exactly with such a simple form for U as we are dealing with, and that a choice has to be made. We make the following choice: galactic orbit and constructed orbit will have exactly the same fundamental points C and D (and because of symmetry C' and D'). We now show that this requirement fixes s and $c_1 - \log c_2$.

Along the curvilinear side CD we have from (11.1)

$$x_2 = R \sin(zs^{-1}) = \text{const},$$

so that

$$\varpi_D - \varpi_C = s \log[\sin(z_D s^{-1}) / \sin(z_C s^{-1})]. \quad (18)$$

Given the points C and D , s can be solved from this equation.

In the points C and D we have (setting again $u=1$ temporarily)

$$U(\varpi, z) = E.$$

Using the abbreviations

$$P = \exp(-\varpi/s), \quad \gamma_1^2 = 1 - a/b = 1 - \gamma^2$$

and using indices for P in the same manner as they occur for ϖ , we find from this condition

$$\begin{aligned} P_0 &= \frac{2P_A - 3P_C}{\gamma_1^2 P_A^2 + \gamma^2 P_C^2 - 2P_C^2} \\ &= \frac{2P_B - 3P_C}{\gamma_1^2 P_B^2 + \gamma^2 P_D^2 - 2P_C^2}. \end{aligned} \quad (19)$$

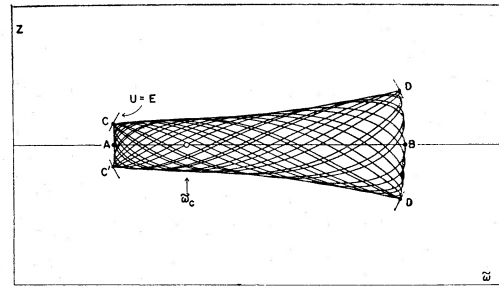


FIG. 2. Region in the meridian plane filled by a galactic box orbit and fundamental points.

These equations can only be satisfied if P_c is a root of the quadratic equation in x

$$\begin{aligned} 4x^2 + 3x \left\{ \gamma_1^2 (P_B + P_A) - \gamma^2 \frac{(P_D + P_C)(P_D - P_C)}{P_B - P_A} \right\} \\ - 2\gamma_1^2 P_A P_B + 2\gamma^2 \frac{P_A P_D^2 - P_B P_C^2}{P_B - P_A} = 0. \end{aligned} \quad (20)$$

P_A and P_B can be expressed in terms of P_C and P_D by observing that along the curvilinear sides AC and BD , respectively,

$$x_1 = e^{c_1} P \cos(z/s) = \text{const}.$$

This gives immediately

$$\begin{aligned} P_A &= P_C \cos(z_C s^{-1}), \\ P_B &= P_D \cos(z_D s^{-1}). \end{aligned} \quad (21)$$

Equations (20) and (21) determine P_c and so ϖ_c . From Eqs. (19) the value of P_0 and so ϖ_0 is obtained. Finally Eq. (16) can be solved for $c_1 - \log c_2$.

CONSTRUCTIONS: ENERGY AND PERIOD

It remains to show how to choose the scaling constant u in such a way that the constructed orbit and the galactic orbit have either exactly the same period or exactly the same maximum kinetic energy. Both requirements cannot be satisfied simultaneously in general as the period and the energy of a constructed orbit are mutually dependent.

We define a characteristic time T_0 (in dimensionless units) of a constructed orbit as the mean time interval of a complete oscillation in the Q_2 coordinate; likewise we define T_g (in kpc-sec/km) of a galactic orbit as the mean time interval of a complete oscillation in z . We speak of the periods T_0 and T_g in referring to these characteristic times; for periodic orbits they are not identical in general to the periods in the usual sense.

The period of a constructed orbit is given by the limit

$$T_0 = \lim_{n \rightarrow \infty} \frac{1}{n} \int_0^{n t_0} \frac{dt}{x_1^2 + x_2^2} \quad (22.1)$$

using the time transformation (6.3) and the form (11.3) for g with which we are concerned. The quantity t_0 is the (constant) period of one oscillation in the x_2 coordinate of the harmonic oscillator; x_1 and x_2 are to be evaluated along the corresponding Lissajous figure in the (x_1, x_2) plane. So

$$x_1 = M \sin(a^{\frac{1}{2}}t + \varphi_1),$$

$$x_2 = N \sin(b^{\frac{1}{2}}t + \varphi_2).$$

The amplitudes M and N expressed in terms of c_1 and the coordinates of the fundamental points C and D in the (ϖ, z) plane are found to be

$$M = \frac{1}{2}e^{c_1}\{P_D \cos(z_D s^{-1}) - P_C \sin(z_C s^{-1})\},$$

$$N = e^{c_1}P_C,$$

using the same definition for P as before. This means that T_0 is proportional to e^{-2c_1} and does not depend on c_2 . Writing

$$T_0 = e^{-2c_1}I \quad (22.2)$$

for (22.1), the integral I is implied in terms of s and the coordinates of C and D by the expressions for x_1 , x_2 , M , and N . Rather than evaluate I by numerical quadrature, we can determine T_0 by numerical integration of the equations of motion of the constructed orbit for an arbitrary value of c_1 and put

$$I = e^{2c_1}T_0.$$

In galactic time units the period is $su^{-1}T_0$; if this is to be a given value T_g , u follows from

$$u = sT_0T_g^{-1} = se^{-2c_1}IT_g^{-1}. \quad (23)$$

We derive a second condition for u from the energy consideration. The largest possible kinetic energy of the constructed orbit is $E - U_c$ (in dimensionless units), which is found to be

$$E - U_c = 2^{-6}ac_2^4\{8\alpha^4 + 20\alpha^2 + (1 + 8\alpha^2)^{\frac{1}{2}} - 1\}.$$

In order that this quantity in galactic units be equal to the largest possible kinetic energy of the given galactic orbit, we have

$$u^2(E - U_c) = E_g.$$

This gives

$$u = [2^3(E_g/a^{\frac{1}{2}}c_2^2)\{8\alpha^4 + 20\alpha^2 + (1 + 8\alpha^2)^{\frac{1}{2}} - 1\}]^{-\frac{1}{2}}. \quad (24)$$

The quantity α is independent of u ; eliminating $c_1 - \log c_2$ from (14) and (16), we obtain

$$\alpha^2 = \frac{1}{8}(4P_c/P_0 - 1)^2 - \frac{1}{8}, \quad (25)$$

so that the value of α depends only on the geometry considered.

Equations (23) and (24) are incompatible in general, as $c_2e^{-c_1}$ is prescribed by the geometry [Eq. (16)]. In the examples discussed in the next section the scaling constant u has been solved from (24), so that the energy

requirement is fulfilled exactly. As a consequence the constructed orbits and the galactic orbits approximated by them, do not have exactly the same period.

RESUME AND EXAMPLES

The successive steps of the construction procedure for a box orbit with given values for the ratio of frequencies γ , the fundamental points C and D and the maximum kinetic energy E_g or the period T_g , can be summarized as follows:

- (1) Determine a and b such that $(a/b)^{\frac{1}{2}} = \gamma$ with $b^{\frac{1}{2}}$ the least positive compatible integer.
- (2) Solve for the scaling constant s from Eq. (18).
- (3) Solve for $x = P_c$ from Eq. (20), and put $\varpi_c = -s \log P_c$; thus the deepest point of the constructed potential well is defined.
- (4) Solve for P_0 from Eq. (19), and put $\varpi_0 = -s \log P_0$; thus the geometrical center of the constructed orbit is defined.
- (5) The fundamental points A and B of the constructed orbit are given by Eqs. (21).
- (6) Find $c_1 - \log c_2$ from Eq. (16).
- (7) Determine α from Eq. (25). Equation (13) gives E for every value of c_2 .
- (8) Evaluate the integral I in (22.2) either by numerical quadrature or by the integration of the equations of motion of the constructed orbit for an arbitrary value of c_1 .
- (9) Solve for u from (23) or (24) with an arbitrary value of c_1 or c_2 , using the given value of T_g or E_g .

In Tables I and II some galactic orbits are compared with approximating orbits constructed in this way. The galactic orbits were obtained by numerical integration using an interpolation formula for representing Schmidt's (1956) model of the mass distribution in the

TABLE I. Galactic orbits. All orbits have the same maximum kinetic energy ($14\,531\text{ km}^2/\text{sec}^2$) and the same angular momentum integral (corresponding to $\varpi_c = 5.2\text{ kpc}$). Units in ϖ and z are kpc; unit in T_g is $0.1\text{ kpc}\cdot\text{sec}\cdot\text{km}^{-1}$.

γ	$b^{\frac{1}{2}}$	ϖ_A	ϖ_B	ϖ_C	z_C	ϖ_D	z_D	T_g
0.51383	2	3.460	10.24	3.455	0.329	10.209	0.779	0.793
0.56728	7	3.520	10.16	3.513	0.450	10.074	1.075	0.880
0.59119	5	3.560	10.13	3.532	0.495	10.000	1.234	0.918
3/5	5	3.573	10.109	3.548	0.513	9.960	1.305	0.939

TABLE II. Constructed orbits. All orbits have the same maximum kinetic energy as in Table I; $c_2 = e$ in all cases. Units in ϖ and $su^{-1}T_0$ as in Table I. Units in s and u are kpc and $10\text{ km}\cdot\text{sec}^{-1}$, respectively.

γ	ϖ_A	ϖ_B	s	ϖ_c	ϖ_0	c_1	u	$su^{-1}T_0$
0.51383	3.462	10.248	7.848	5.115	6.143	1.783	4.932	0.874
0.56728	3.526	10.151	7.558	5.091	6.135	1.812	1.228	0.974
0.59119	3.549	10.108	7.114	5.031	6.098	1.857	1.550	1.044
3/5	3.567	10.084	6.905	5.008	6.084	1.881	1.620	1.081

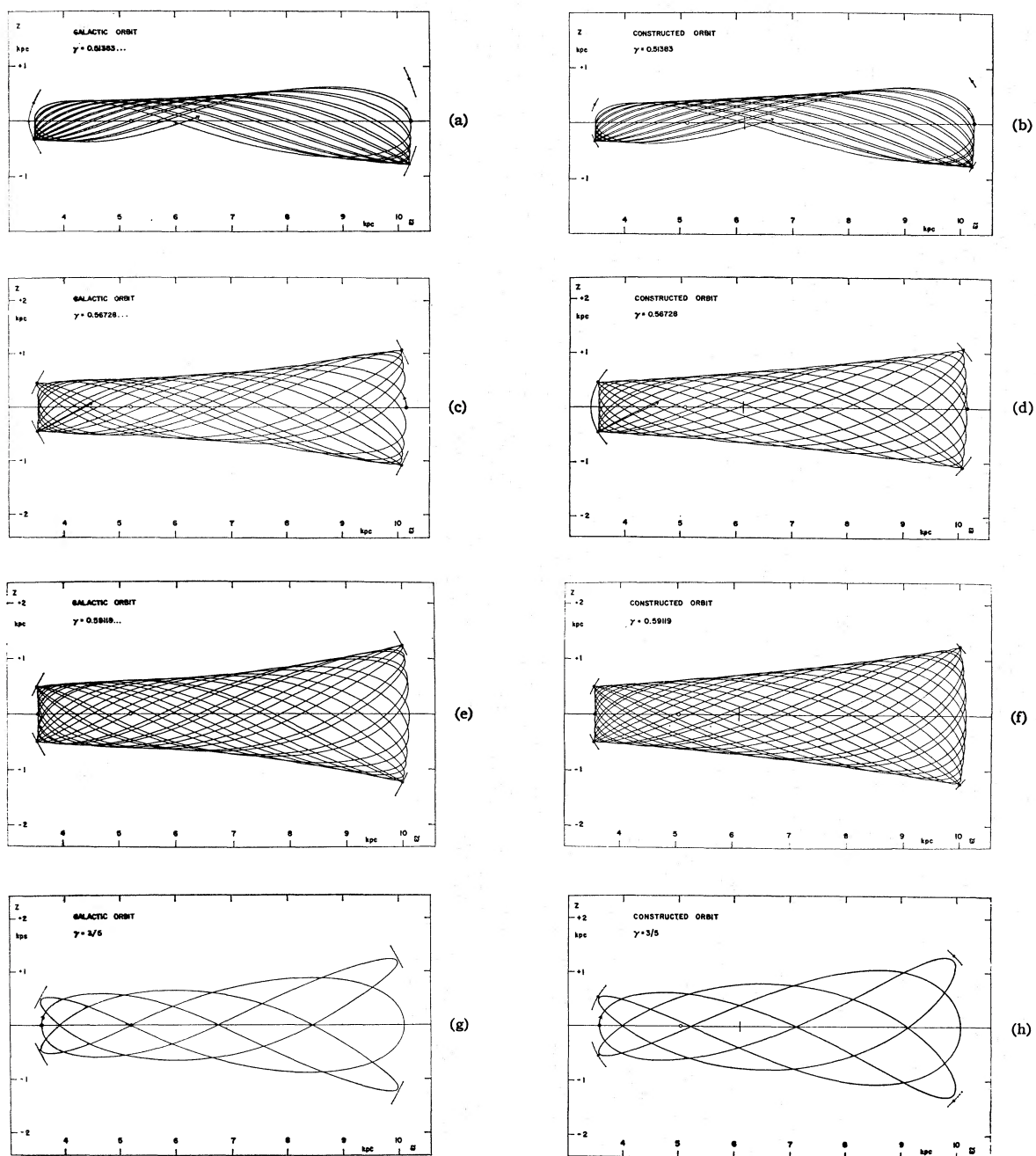


FIG. 3. Comparison of galactic orbits and constructed orbits approximating them. An open circle marks the point of lowest potential energy; filled circles mark fundamental points. Galactic and corresponding constructed orbit are shown in the same phase.

Galaxy; see also Ollongren (1962). All galactic orbits have the same value of the angular momentum integral and the maximum kinetic energy, but on the average most of the latter is present in radial motion rather than in motion perpendicular to the plane. Iso-energetic orbits with z amplitudes larger than about 1.5 kpc cannot be treated in the same way, because there is numerical evidence that the side CD of the region filled by a box orbit develops a minimum with respect

to z . It is not possible to construct orbits with this property using the particular similarity transformation discussed above.

In Fig. 3 the same galactic and the constructed orbits are shown in the (ϖ, z) coordinate system. The figures were drawn by computer following straightforward numerical integration of the equations of motion. As initial conditions $z=0$, $p_z=0$, $E_0=14\,531$ km²/sec², ϖ_A or ϖ_B were chosen.

CONCLUSION

In the foregoing we have applied a particular similarity transformation to harmonic oscillators with given ratios of the oscillation frequencies. The transformation and the parameters figuring in it were tailored so as to construct orbits which have certain properties in common with numerically computed galactic stellar orbits. An exact match is not obtained, but the constructed orbits may be used as reference orbits for further analysis. A justification for doing this is provided by the fact that it becomes possible to take into account the variability of periods of z oscillations for galactic orbits; reference orbits constructed by means of canonical point transformations leading to a separable system do not give this possibility.

Tables I and II indicate that only galactic orbits with amplitudes less than about 1 kpc can be approximated reasonably well by constructions of the kind treated here. It appears that there are several directions in which an improvement of this situation can be obtained:

(a) More than one similarity transformation may be applied to a harmonic oscillator. Since transformations of this type form a (mathematical) equivalence class, the result is again a similarity transformation. Thus more parameters can be introduced in the construction procedure; the latter may, of course, become more difficult owing to increased complexity in the transformed Hamiltonian.

(b) Other metrics than the one considered above may be introduced in the (x_1, x_2) configuration space. If a solution of the partial differential equations for

f_1 and f_2 can be found, other forms for the resulting Hamiltonian are obtained as a result.

It is hoped that it will be possible to find in this way a useful representation of galactic stellar orbits evidently similar to harmonic oscillators, but with higher Z velocities than treated here. Such orbits are known from previous numerical work, but had to be excluded from the present treatment.

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